

Name of student: Dhiraj Suresh patil

PRN: 231106024

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import re

def create_playfair_matrix(key):
    key = key.upper()
    key = key.replace(" ", "")
    key = re.sub(r'^[A-Z]', '', key) # Remove non-alphabetic characters
    key = key.replace("J", "I")
    matrix = []
    chars = []
    for char in key:
        if char not in chars:
            chars.append(char)
    for i in range(65, 91):
        char = chr(i)
        if char not in chars:
            if char == 'J':
                if 'I' not in chars:
                    chars.append('I')
            else:
                chars.append(char)

    for i in range(0, 25, 5):
        matrix.append(chars[i:i+5])
    return matrix

def get_char_indices(matrix, char):
    for i, row in enumerate(matrix):
        try:
            j = row.index(char)
            return i, j
        except ValueError:
            pass
    return None, None

def encrypt_pair(matrix, char1, char2):
    row1, col1 = get_char_indices(matrix, char1)
    row2, col2 = get_char_indices(matrix, char2)

    if row1 == row2:
        return matrix[row1][(col1 + 1) % 5], matrix[row2][(col2 + 1) % 5]
    elif col1 == col2:
        return matrix[(row1 + 1) % 5][col1], matrix[(row2 + 1) % 5][col2]
    else:
        return matrix[row1][col2], matrix[row2][col1]

def decrypt_pair(matrix, char1, char2):
    row1, col1 = get_char_indices(matrix, char1)
    row2, col2 = get_char_indices(matrix, char2)

    if row1 == row2:
        return matrix[row1][(col1 - 1) % 5], matrix[row2][(col2 - 1) % 5]
    elif col1 == col2:
        return matrix[(row1 - 1) % 5][col1], matrix[(row2 - 1) % 5][col2]
    else:
        return matrix[row1][col2], matrix[row2][col1]

def playfair_encrypt(plaintext, key):
    matrix = create_playfair_matrix(key)
    plaintext = plaintext.upper()
    plaintext = re.sub(r'^[A-Z]', '', plaintext)
    plaintext = plaintext.replace("J", "I")

    if len(plaintext) % 2 != 0:
        plaintext += "X"

    ciphertext = ""
    for i in range(0, len(plaintext), 2):
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        char1 = plaintext[i]
        char2 = plaintext[i+1]
        if char1 == char2:
            char2 = 'X'
            i -= 1

        c1, c2 = encrypt_pair(matrix, char1, char2)
        ciphertext += c1 + c2
    return ciphertext

def playfair_decrypt(ciphertext, key):
    matrix = create_playfair_matrix(key)
    ciphertext = ciphertext.upper()
    ciphertext = re.sub(r'^[A-Z]', '', ciphertext)
    ciphertext = ciphertext.replace("J", "I")

    plaintext = ""
    for i in range(0, len(ciphertext), 2):
        char1 = ciphertext[i]
        char2 = ciphertext[i+1]
        p1, p2 = decrypt_pair(matrix, char1, char2)
        plaintext += p1 + p2
    return plaintext

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# Example usage (Sample Playfair key: Monarchy and plaintext: instrument)
key = input("Enter the Playfair key: ")
plaintext = input("Enter the plaintext: ")
ciphertext = playfair_encrypt(plaintext, key)
print("Ciphertext:", ciphertext)

decrypted_text = playfair_decrypt(ciphertext, key)
print("Decrypted text:", decrypted_text)

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Enter the Playfair key: MONARCHY
Enter the plaintext: RCPITSHIRPUR
Ciphertext: MDSFLTBFOTZM
Decrypted text: RCPITSHIRPUR

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